Supplementary material: Qualifying Image-Grounded Tag Retrieval in Piping and Instrumentation Diagrams with Source-Isolated Counterfactual Evaluation

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S1. Positioning and evidence contract

The manuscript evaluates a bounded engineering decision: whether a specified model and visual-input budget support candidate-tag retrieval on unseen source drawings. It does not propose a new recognition architecture or claim complete P&ID digitization. Table S1 distinguishes the contribution from adjacent work using only dimensions relevant to that decision.

Table 1: Positioning relative to representative P&ID and document-VQA work. Bibliographic details and DOIs appear in the main manuscript.

Work	Primary input/domain	Primary output	Evaluation emphasis	Relation to this study
Paliwal et al. (2021)	Synthetic P&ID images	Digitized diagram elements	End-to-end digitization pipeline	Source resource and broader digitization target
Kim et al. (2021)	High-density industrial P&ID images	Symbols and text	Industrial recognition performance	Demonstrates specialized recognition beyond generic VLM prompting
Su et al. (2024)	High-resolution industrial P&ID images	Symbols, text, and pipelines	Multi-stage visual recognition	Motivates task-specific rather than aggregate evaluation
Gupta et al. (2025)	500 Dataset-PID sheets	64,000 graph-derived QA records	P&ID question answering	Dataset and task taxonomy used here
Nourbakhsh et al. (2025)	Document VQA	Grounding-aware evaluation	Whether outputs are attributable to documents	Adjacent grounding motivation
This study	Source-isolated PIDQA drawings	Candidate value-tag sets	Correct, shuffled, and absent images; budget and scope gates	Evaluation and qualification workflow

The evidence phases are:

1. configuration records, used to freeze prompt, parser, legal tag grammar, and inference conditions;
2. primary Set B qualification, using 100 unseen source drawings;

3. descriptive Set B fusion, where the component predictions motivated a complete four-rule operating family;
4. frozen-rule within-family checks on seed-29/31 partitions, followed by explicit source-overlap exclusions.

The last phase is stability evidence inside PIDQA, not an external replication.

S2. Frozen prompts, parsing, and answer isolation

The exact Qwen P0 template is:

```
You analyze a piping and instrumentation diagram (P&ID). Answer only from the
visible diagram.  Return only the final answer, without explanation or Markdown.
Question:  {question}
```

The matched InternVL prompt uses the same sentence sequence. Image conditions prepend the model's `<image>` placeholder; the no-image condition does not. All generation is greedy. Qwen rows record maximum image side and output cap separately. InternVL image rows record realised 448-pixel tiles and output cap. The tokenizer-corrected InternVL pass sets `fix_mistral_regex=true`; the warned diagnostic pass is retained in the archive but is not used for reported scores.

Strict value scoring lowercases, trims, de-duplicates, sorts, and compares tag sets. The conservative grammar accepts alphabetic prefixes followed by numeric, hyphen-numeric, or space-numeric suffixes (for example `KL-58999`, `UV-00-001`, `SDL 299`, and `CV1`); bare uppercase tags are accepted after a frozen stopword filter. Empty strings and `[]` map to an empty set. The parser never reads the hidden reference.

The answer-isolation audit covers seven public manifests and 13 primary E2–E8 raw output files. Every successful output's legacy `answer` field aliases generated `raw` text rather than a reference label. Scorer- only references are used after inference for evaluation and for no prediction construction.

S3. Estimator and uncertainty definitions

For source i, let T_i and P_i be reference and prediction sets. The element counts are

$$TP_i = |T_i \cap P_i|, \qquad FP_i = |P_i \setminus T_i|, \qquad FN_i = |T_i \setminus P_i|.$$

Micro precision and recall pool the counts across sources, and pooled F1 is

$$F1_{\text{micro}} = \frac{2\sum_i TP_i}{2\sum_i TP_i + \sum_i FP_i + \sum_i FN_i}.$$

Source-macro quantities compute precision, recall, and F1 separately for each source and average each metric over sources. Exact-set accuracy is the source mean of the indicator that $T_i = P_i$.

When $T_i = P_i = \emptyset$, the per-source precision, recall, and F1 are set to 1 because the source is exactly resolved. When $T_i \neq \emptyset$ and $P_i = \emptyset$, all three are 0. Seven of 100 Set B value sources have empty references. To make the convention visible, all-source macro and non-empty-reference macro values are reported together.

Intervals are 95% percentile intervals from 10,000 source bootstrap replicates. Paired comparisons resample the same source indices for both conditions. Random numbers use `numpy.random.default_rng` (PCG64), with fixed comparison-specific seeds recorded in `rineng_revision_analysis_v6.json`. No record-level interval and no post-observation equivalence margin are used.

S4. Source splitting and aggregate boundary

The input-retrieval-plus-prior diagnostic uses exact image SHA-256 plus a released-template semantic signature and a training-only task-majority fallback. It uses no learned embedding, nearest-neighbour search, or `source_id` cache key. Each split has 38,400 training and 12,800 test questions; the 20% calibration partition is unused.

Table 2: All five pre-specified split diagnostics.

Seed	Question-random	Source-isolated	Difference	Difference (points)
3	0.5199	0.3128	+0.2071	+20.71
17	0.5253	0.3121	+0.2132	+21.32
29	0.5248	0.3148	+0.2101	+21.01
43	0.5217	0.3141	+0.2077	+20.77
71	0.4554	0.3194	+0.1360	+13.60
Mean	0.5094	0.3146	+0.1948	+19.48

The independent training-only task-majority diagnostic scores 0.3375 aggregate strict accuracy on Set B. Correct-image Qwen aggregate strict accuracy is 0.2875 and no-image accuracy is 0.3025. These are not tag F1 values; they are reported to show why aggregate benchmark accuracy does not qualify image- grounded retrieval.

S5. Qwen counterfactual and budget controls

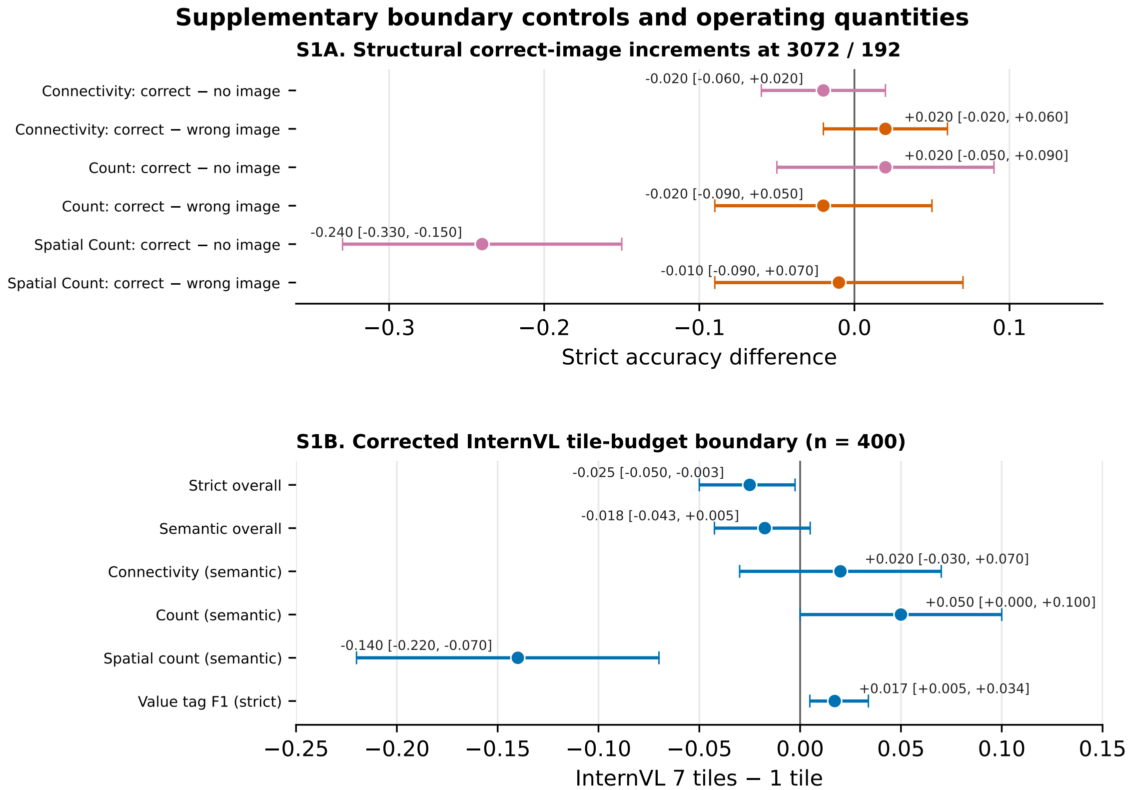
Table 3: Value-tag controls. Effects are paired source-level pooled-F1 differences with 10,000 source-bootstrap replicates.

Contrast	Baseline F1	Condition F1	Difference [95% interval]
3072 minus 768, 192-token cap	0.0000	0.5549	+0.5549 [0.4774, 0.6314]
3072 minus 768, 512-token cap	0.0003	0.5253	+0.5250 [0.4224, 0.6293]
Correct minus source-shuffled, 3072/192	0.0062	0.5549	+0.5487 [0.4729, 0.6239]
Correct minus no image, 3072/192	0.0000	0.5549	+0.5549 [0.4799, 0.6322]

Table 4: Recorded operating quantities. Tensor elements are processor outputs, not asserted visual-token counts. Peak memory is observed allocated memory on the H200 run.

Condition	Actual input	Cap	Tokens mean	Tokens p95	Cap rate	Peak GiB
Qwen 768 correct, value	2,211,840 elements	512	421.2	512	79.0%	16.50
Qwen 3072 correct, value	35,979,264 elements	512	54.7	512	6.0%	18.17
InternVL one tile, all tasks	1 / 602,112 elements	192	51.0	192	17.8%	16.04
InternVL seven tiles, all tasks	7 / 4,214,784 elements	192	16.4	63	2.8%	17.41

For the matched Qwen comparison, the processor-element ratio is 16.27. The 768-side condition generates substantially longer text and reaches the cap more often. End-to-end latency is therefore not interpreted as isolated visual-encoder cost. Legacy 192-token Qwen rows lack processor-element and peak-allocation telemetry; these values are not back-filled.



S1C. Recorded operating quantities

Condition	Actual input	Tokens mean / p95	Cap rate	Latency s mean / p95	Peak GiB
Qwen 768 correct, value, 512 cap	2211840	421.2 / 512	79.0%	9.21 / 11.26	16.50
Qwen 3072 correct, value, 512 cap	35979264	54.7 / 512	6.0%	1.78 / 11.68	18.17
InternVL 1 tile, all tasks, 192 cap	1 / 602112	51.0 / 192	17.8%	1.42 / 5.22	16.04
InternVL 7 tiles, all tasks, 192 cap	7 / 4214784	16.4 / 63	2.8%	0.61 / 1.86	17.41

Latency is end-to-end generation time and co-varies with output length; it is not interpreted as visual-encoding cost. NR = not recorded.

Figure 1: Structural correct-image increments, corrected InternVL tile-budget boundary, and recorded operating quantities. This boundary figure is retained in the supplement so that the main tag-retrieval claim is not generalized to all tasks or model families.

S6. OCR literal-versus-join ablation

PaddleOCR processes the complete source drawing with the English PP-OCRv3 detector and PP-OCRv4 recognizer, CPU execution, no crop, no angle classifier, and detector maximum side 3072. The geometry rule joins a recognized alphabetic prefix to the nearest vertically stacked numeric fragment within frozen distance limits. Fully formed tags with final numeric segments longer than two characters are never extended. Coordinates and grammar are used; references are not.

Table 5: OCR parsing ablation on the same 100 Set B value sources.

Parse	TP	FP	FN	Precision	Recall	Micro F1	Macro F1	Exact
Literal	116	44	232	0.7250	0.3333	0.4567	0.4227	0.1200
Geometry join v1	159	29	189	0.8457	0.4569	0.5933	0.5618	0.2100

There are 45 join operations. Relative to literal parsing, the joined output adds 44 true tags and one false candidate, removes 16 false candidates and one true tag, improves per-source F1 on 16 drawings, degrades it on one, and is unchanged on 83. This is a deterministic normalization ablation; it is not a claim that every joined string is semantically correct outside the benchmark grammar.

S7. Complete operating family, estimators, and workload

All rules use the fixed Qwen and geometry-joined OCR predicted sets only. No reference, fitted confidence threshold, or rule re-selection enters a fused prediction.

Table 6: Set B components and complete four-rule family. The non-empty macro column averages only the 93 sources containing at least one reference tag.

Mode	TP	FP	FN	P	R	Micro F1	Macro F1	Non-empty macro F1	Exact	Empty predictions
Qwen	192	152	156	0.5581	0.5517	0.5549	0.5810	0.6139	0.1900	1
OCR geometry	159	29	189	0.8457	0.4569	0.5933	0.5618	0.5396	0.2100	18
Set union	245	180	103	0.5765	0.7040	0.6339	0.6602	0.6991	0.1900	1
Set intersection	106	1	242	0.9907	0.3046	0.4659	0.4312	0.3992	0.1700	41
OCR if non-empty, else Qwen	175	39	173	0.8178	0.5029	0.6228	0.6073	0.6423	0.2200	1
Qwen if non-empty, else OCR	192	152	156	0.5581	0.5517	0.5549	0.5810	0.6139	0.1900	1

Table 7: Per-drawing candidate workload, median [IQR]. False candidates are predicted tags absent from the benchmark reference.

Mode	Candidate count	False-candidate count
Qwen	3 [1, 4]	0 [0, 2]
OCR literal	1 [1, 2]	0 [0, 1]
OCR geometry	2 [1, 3]	0 [0, 0]
Set union	4 [2, 5]	1 [0, 2]
Set intersection	1 [0, 2]	0 [0, 0]
OCR if non-empty, else Qwen	2 [1, 3]	0 [0, 1]

Table 8: Selected paired differences on Set B across estimators.

Comparison	Estimator	F1 difference	95% interval
Union minus Qwen	Micro pooled	+0.0790	[0.0372, 0.1261]
Union minus Qwen	Source macro	+0.0792	[0.0373, 0.1250]
Union minus Qwen	Non-empty source macro	+0.0852	[0.0398, 0.1346]
Union minus OCR	Micro pooled	+0.0406	[-0.0282, 0.1118]
Union minus OCR	Source macro	+0.0984	[0.0130, 0.1819]
OCR-first minus OCR	Micro pooled	+0.0295	[0.0051, 0.0590]
Intersection minus Qwen	Micro pooled	-0.0890	[-0.1616, -0.0181]

Intersection is not intended to maximize F1; it exchanges coverage for a shortlist with precision 0.9907 and source-bootstrap interval [0.9681, 1.0000]. The estimator dependence of union minus OCR is retained rather than collapsed into a single favourable number.

S7.1 Cost-sensitive operating-mode rule

With $C_{FP} = 1$ and $r = C_{FN}/C_{FP}$, loss is $L = rFN + FP$. The following point-estimate intervals are the exact lower envelope of the Set B count vectors. Adjacent methods tie at each finite boundary.

Table 9: Exact Set B cost-sensitive decision rule.

Cost ratio r	Selected mode	Loss line	Engineering role
$0 \le r < 0.5283$	Intersection	$242r + 1$	Precision-first shortlist
$0.5283 < r < 0.6250$	Joined OCR	$189r + 29$	Low-false-positive OCR
$0.6250 < r < 2.0143$	OCR-first	$173r + 39$	Balanced fallback
$r > 2.0143$	Union	$103r + 180$	Recall-first coverage

Table 10: Descriptive source-bootstrap uncertainty of adjacent switch ratios.

Adjacent modes	Point switch	Bootstrap median	95% interval
Intersection / joined OCR	0.5283	0.5323	[0.2933, 0.9584]
Joined OCR / OCR-first	0.6250	0.6111	[0.1765, 1.7500]
OCR-first / union	2.0143	2.0000	[1.2000, 3.3231]

The first and third switch ratios are positive in all 10,000 resamples; the middle ratio is positive in 9,994. These are uncertainty summaries for a sample-derived operating rule, not elicited real-world costs. The complete 321-point loss and probability grid is distributed in `reports/generated/rineng_cost_sensitive_operating_modes_v8.csv`.

S8. Source membership audit and frozen-rule checks

Set B shares 17 sources with seed 29 and 17 with seed 31. Seed 29 and seed 31 share 20 sources; two sources occur in all three sets. Set B exclusion leaves 83 sources in each check, with 18 sources still common to the two checks. A second exclusion creates two 65-source subsets that share no source with Set B or with one another.

Table 11: Union rule across original, Set-B-excluded, and mutually disjoint subsets. Original-partition rows reproduce the earlier descriptive scorer; new exclusion rows use the v6 estimator.

Subset	n	Qwen micro F1	Union micro F1	Union–Qwen [95% interval]
Original seed 29	100	0.6783	0.7167	+0.0384 [0.0059, 0.0740]
Original seed 31	100	0.6500	0.6918	+0.0418 [0.0048, 0.0852]
Seed 29 excluding Set B	83	0.6717	0.7083	+0.0366 [0.0002, 0.0751]
Seed 31 excluding Set B	83	0.6417	0.6856	+0.0440 [0.0021, 0.0920]
Seed 29 strictly disjoint	65	0.6587	0.6911	+0.0325 [-0.0078, 0.0764]
Seed 31 strictly disjoint	65	0.6158	0.6667	+0.0509 [0.0012, 0.1093]

Table 12: Source-macro sensitivity after overlap exclusions.

Subset	Qwen macro F1	Union macro F1	Difference [95% interval]
Seed 29 excluding Set B	0.6643	0.7094	+0.0451 [0.0043, 0.0892]
Seed 31 excluding Set B	0.6416	0.6789	+0.0373 [-0.0082, 0.0854]
Seed 29 strictly disjoint	0.6467	0.6846	+0.0379 [-0.0088, 0.0885]
Seed 31 strictly disjoint	0.6049	0.6474	+0.0425 [-0.0123, 0.0974]

All point estimates are positive, but interval exclusion of zero is not uniform. These rows support direction and uncertainty reporting only. They do not support an independent-dataset or cross-family statement.

S9. Deterministic complementarity and error taxonomy

Across the 348 Set B reference tags, 106 are recovered by both Qwen and joined OCR, 86 by Qwen only, 53 by OCR only, and 103 by neither. Among false candidates in the union, one is produced by both instruments, 151 by Qwen only, and 28 by OCR only. At drawing level, Qwen contributes at least one unique true tag on 43 sources, OCR on 17, both on seven, and neither on 33.

Table 13: Mutually exclusive source-level outcome patterns. Empty prediction means an empty set against a non-empty reference; exact empty–empty cases are counted as exact.

Mode	Exact	Complete + false	Partial clean	Partial + false	False only	Empty
Qwen	19	9	32	19	21	0
OCR literal	12	1	36	14	25	12
OCR geometry	21	1	42	15	9	12
Set union	19	16	22	33	10	0
Set intersection	17	0	47	0	1	35
OCR-first fallback	22	1	46	16	15	0

A separate deterministic string-form diagnostic pairs false candidates and missed tags by compact edit form. For Qwen it counts 40 prefix/suffix fragments, nine single-character edits, three other paired string differences, 100 unpaired false candidates, and 104 unpaired misses. Joined OCR counts five prefix/suffix fragments, two other paired differences, 22 unpaired false candidates, and 182 unpaired misses. Literal OCR has 20 prefix/suffix fragments. These are reproducible form categories, not manual failure annotations and not causal explanations.

Table 14 gives a three-case boundary gallery without calling the cases representative. Before selection, the Qwen strata were fixed as success (exact set), partial recovery with at least one false candidate, and failure with false candidates but no recovered tag. The eligible strata contain 19, 19, and 21 sources. Within each stratum we selected the source with the minimum SHA-256 of the frozen analysis version, stratum label, and `source_id`. This deterministic, outcome-conditioned selection is an audit display only; it did not construct or tune any prediction.

Table 14: Deterministically selected success, partial, and failure boundary cases. Tags are the normalized sets used by the scorer. Joined OCR is the frozen full-image geometry rule.

Qwen stratum	Source and reference	Qwen candidates	Joined OCR candidates	Boundary shown
Success (19 eligible)	sheet-282: ro-10 364; ro-10 525; ro-10 545	Same three tags	Same three tags	Both methods and both set rules are exact.
Partial (19 eligible)	sheet-191: cd-15150; cd-58014	cd-58014; cd-81354	cd-58014	Intersection removes the false candidate but cannot recover the shared miss.
Failure (21 eligible)	sheet-491: sdl 333; sdl 343; sdl 658; sdl 871	Hyphenated variants sdl-333; sdl-343; sdl-658; sdl-871	Exact four reference tags	Frozen normalization treats space and hyphen forms differently: union recovers all four but retains four false forms; intersection is empty.

S10. Cross-model and semantic-boundary controls

InternVL3.5-8B uses the same 100 value records and source derangement as the Qwen grounding control. Correct and source-shuffled image conditions each realise seven tiles (4,214,784 tensor elements); no-image passes no pixel tensor. All use a 512-token cap.

Table 15: Corrected cross-model boundary and frozen OCR comparator.

Condition	Precision	Recall	F1	Exact	Mean tokens	Mean latency (s)
InternVL correct image	0.0091	0.0201	0.0126	0.0000	61.530	1.798
InternVL source-shuffled image	0.0000	0.0000	0.0000	0.0000	70.750	2.054
InternVL no image	0.0000	0.0000	0.0000	0.0000	187.410	5.518
PaddleOCR literal	0.7250	0.3333	0.4567	0.1200	n/a	12.446
PaddleOCR geometry join	0.8457	0.4569	0.5933	0.2100	n/a	12.446

InternVL correct-minus-source-shuffled and correct-minus-no-image F1 effects are both +0.0126, with intervals [0.0035, 0.0261] and [0.0035, 0.0262]. The direction is detected but the magnitude is not a practical replication of the Qwen operating point.

S10.1 Frozen three-model, three-subset, two-prompt matrix

The V7 extension was frozen before inference and contains 54 cells: three models, three pairwise source-disjoint PIDQA subsets, two pre-existing prompts, and correct, source-shuffled, and no-image conditions. Every cell covers all four PIDQA tasks. The 16,560 prediction rows contain no inference error, no duplicate instance identifier within a cell, and no true `test_answer_used` flag. All rows carry the same frozen-plan SHA-256.

Table 16 reports all 36 primary correct-minus-control value-tag contrasts. Every 95% paired source-bootstrap interval has a positive lower bound. Qwen3-VL-8B and Qwen3-VL-32B show correct-image F1 ranges of 0.5253–0.6587 and 0.5697–0.7361; InternVL3.5-8B remains at 0.0090–0.0257 under the native, at-most-12-tile setting. Thus the requested-drawing direction is repeatable within PIDQA, while magnitude at that stage remains visual-budget and model-family dependent. Section S13 reports the subsequent closest-safe 54-tile qualification.

Table 16: Frozen V7 value-tag counterfactual matrix. Effects are correct-image minus control pooled strict tag F1; brackets are 95% percentile intervals from 10,000 paired source bootstrap replicates.

Model	Subset	Prompt	Correct F1	Shuffled F1 / effect [CI]	No-image F1 / effect [CI]
Qwen3-VL-8B	Set B (100)	P0	0.525	0.006 / +0.519 [+0.414, +0.621]	0.000 / +0.525 [+0.424, +0.628]
Qwen3-VL-8B	Set B (100)	P1	0.593	0.003 / +0.591 [+0.517, +0.663]	0.000 / +0.593 [+0.519, +0.666]
Qwen3-VL-8B	Seed 29 (65)	P0	0.659	0.015 / +0.644 [+0.563, +0.721]	0.000 / +0.659 [+0.580, +0.734]
Qwen3-VL-8B	Seed 29 (65)	P1	0.656	0.015 / +0.641 [+0.564, +0.716]	0.000 / +0.656 [+0.578, +0.731]
Qwen3-VL-8B	Seed 31 (65)	P0	0.616	0.025 / +0.591 [+0.478, +0.697]	0.000 / +0.616 [+0.512, +0.718]
Qwen3-VL-8B	Seed 31 (65)	P1	0.630	0.025 / +0.605 [+0.497, +0.713]	0.000 / +0.630 [+0.524, +0.736]
Qwen3-VL-32B	Set B (100)	P0	0.570	0.006 / +0.564 [+0.438, +0.702]	0.000 / +0.570 [+0.445, +0.706]
Qwen3-VL-32B	Set B (100)	P1	0.614	0.005 / +0.609 [+0.492, +0.717]	0.000 / +0.614 [+0.497, +0.721]
Qwen3-VL-32B	Seed 29 (65)	P0	0.736	0.008 / +0.728 [+0.653, +0.800]	0.000 / +0.736 [+0.664, +0.806]
Qwen3-VL-32B	Seed 29 (65)	P1	0.687	0.009 / +0.678 [+0.603, +0.755]	0.000 / +0.687 [+0.612, +0.764]
Qwen3-VL-32B	Seed 31 (65)	P0	0.657	0.022 / +0.635 [+0.527, +0.736]	0.000 / +0.657 [+0.553, +0.755]
Qwen3-VL-32B	Seed 31 (65)	P1	0.645	0.021 / +0.624 [+0.521, +0.721]	0.000 / +0.645 [+0.544, +0.744]
InternVL3.5-8B	Set B (100)	P0	0.013	0.000 / +0.013 [+0.003, +0.027]	0.000 / +0.013 [+0.004, +0.026]
InternVL3.5-8B	Set B (100)	P1	0.009	0.000 / +0.009 [+0.003, +0.018]	0.000 / +0.009 [+0.003, +0.018]
InternVL3.5-8B	Seed 29 (65)	P0	0.012	0.000 / +0.012 [+0.002, +0.031]	0.000 / +0.012 [+0.002, +0.031]
InternVL3.5-8B	Seed 29 (65)	P1	0.012	0.000 / +0.012 [+0.003, +0.027]	0.000 / +0.012 [+0.003, +0.026]
InternVL3.5-8B	Seed 31 (65)	P0	0.026	0.002 / +0.024 [+0.008, +0.051]	0.000 / +0.026 [+0.010, +0.053]
InternVL3.5-8B	Seed 31 (65)	P1	0.020	0.001 / +0.019 [+0.007, +0.040]	0.000 / +0.020 [+0.007, +0.042]

Figure 2 reports every P1-minus-P0 correct-image contrast. Eight of nine intervals include zero. The only exception is Qwen3-VL-32B on seed 29, where P1 is lower by 0.0487 with interval [-0.1044, -0.0025]. No best-prompt selection is applied.

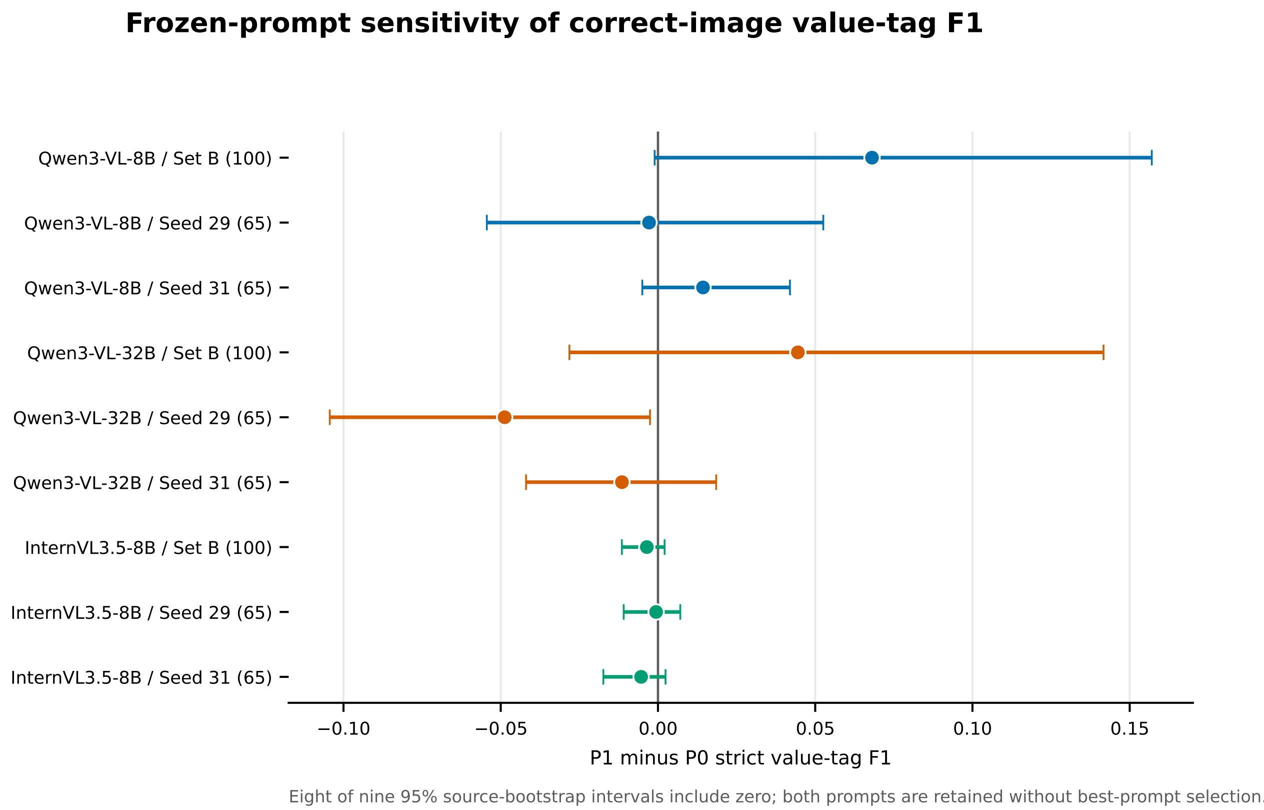


Figure 2: Frozen-prompt sensitivity of correct-image strict value-tag F1. Whiskers are 95% intervals from 10,000 paired source-bootstrap replicates.

Table 17 preserves the all-task boundary: exact and aggregate accuracy must not be substituted for the pooled tag-set F1 estimator used by the qualified retrieval claim.

Table 17: Correct-image strict accuracy by task under the frozen V7 configurations. Value F1 is pooled strict value-tag F1 and is not interchangeable with exact-set accuracy.

Model	Subset	Prompt	Overall	Connectivity	Count	Spatial count	Value exact	Value F1
Qwen3-VL-8B	Set B (100)	P0	0.287	0.560	0.100	0.300	0.190	0.525
Qwen3-VL-8B	Set B (100)	P1	0.295	0.570	0.110	0.310	0.190	0.593
Qwen3-VL-8B	Seed 29 (65)	P0	0.315	0.569	0.108	0.323	0.262	0.659
Qwen3-VL-8B	Seed 29 (65)	P1	0.273	0.554	0.092	0.246	0.200	0.656
Qwen3-VL-8B	Seed 31 (65)	P0	0.258	0.523	0.046	0.215	0.246	0.616
Qwen3-VL-8B	Seed 31 (65)	P1	0.265	0.538	0.062	0.215	0.246	0.630
Qwen3-VL-32B	Set B (100)	P0	0.367	0.570	0.070	0.530	0.300	0.570
Qwen3-VL-32B	Set B (100)	P1	0.365	0.570	0.070	0.530	0.290	0.614
Qwen3-VL-32B	Seed 29 (65)	P0	0.396	0.554	0.062	0.554	0.415	0.736
Qwen3-VL-32B	Seed 29 (65)	P1	0.388	0.554	0.062	0.554	0.385	0.687
Qwen3-VL-32B	Seed 31 (65)	P0	0.327	0.523	0.062	0.446	0.277	0.657
Qwen3-VL-32B	Seed 31 (65)	P1	0.327	0.523	0.062	0.446	0.277	0.645
InternVL3.5-8B	Set B (100)	P0	0.033	0.000	0.110	0.020	0.000	0.013
InternVL3.5-8B	Set B (100)	P1	0.080	0.150	0.140	0.030	0.000	0.009
InternVL3.5-8B	Seed 29 (65)	P0	0.046	0.000	0.077	0.092	0.015	0.012
InternVL3.5-8B	Seed 29 (65)	P1	0.058	0.092	0.077	0.046	0.015	0.012
InternVL3.5-8B	Seed 31 (65)	P0	0.031	0.000	0.046	0.062	0.015	0.026
InternVL3.5-8B	Seed 31 (65)	P1	0.058	0.123	0.077	0.015	0.015	0.020

The symbol-legend control preserves prototype crops, grid, font, dimensions, and second-image count. Visible legend minus raw spatial-count accuracy is +0.3100 [0.2200, 0.4100] at 768-side and

+0.2400 [0.1500, 0.3400] at 3072-side. Correct numeric labels minus a layout-matched cyclic permutation is +0.0000 [-0.0400, 0.0400] at both budgets. The visual context affects the task, but the tested numeric mapping semantics do not show a detected effect.

S11. Artifact provenance and deterministic reproduction

Run

```
python scripts/reproduce_submission_v6.py -root .
```

from the repository root. The script re-scores immutable outputs, rebuilds the v6 analysis and figures, runs unit and package checks, compiles manuscript and supplement PDFs, and performs page-render validation. It does not rerun model inference or retry the unavailable external-data branch.

The recovered V7 matrix is independently reproduced without new inference by running

```
python scripts/reproduce_rineng_overnight_v7.py -root .
```

This command recomputes all 54 cell metrics, 36 paired effects and intervals, and nine prompt-sensitivity intervals before rebuilding the V7 tables, figures, and SHA-256 artifact manifest.

The principal v6 and V7 machine-readable artifacts are:

- `reports/generated/rineng_revision_analysis_v6.json`;
- `reports/generated/rineng_revision_analysis_v6.csv`;
- `reports/generated/rineng_revision_per_source_v6.csv`;
- `reports/generated/rineng_revision_error_taxonomy_v6.csv`;
- `paper/figures/figure_metadata_v6.json`;
- `reports/generated/rineng_overnight_v7_score.json`;
- `reports/generated/rineng_overnight_v7_validation.json`;
- `reports/RINENG_OVERNIGHT_V7_ARTIFACT_MANIFEST.json`;
- `paper/figures/figure_metadata_v7.json`.

The analysis JSON records the seven immutable input paths, byte sizes, and SHA-256 hashes. The release archive additionally records dependency versions, source manifests, raw model output identifiers and revisions, tests, and a top-level SHA-256 inventory. GPU inference used NVIDIA H200 devices; model weights are not redistributed.

The public release excludes author-side editorial prompts, model-review text, private working notes, and submitter identity fields because none is required to reproduce a result. Scorer-only references are included with explicit separation from model-facing manifests. The public DOI/URL, author identities, affiliations, CRediT roles, funding, competing-interest statement, and license choice remain submitter-owned administrative fields.

S12. External-data provenance and task-fit boundary

The DEXPI external-family branch freezes the official Public Example PIDs repository at commit `a23d61e2e089eb2ca464cd552f9ae580a2785963`. The byte-preserved CC BY 4.0 license has SHA-256 `7e7170e3cebf88a9f60c7b8421418323c09304da1af4d5e90f4da1dc1c8a2661`. Among 86 exact same-directory image/XML stem pairs, an automated audit retains a source only when a structured XML tag value also occurs in an XML graphic text declaration. Duplicate image hashes are removed; one image per logical test case is selected before SHA-ranked vendor variants. The frozen result is 35 images from 26 logical cases and 65 prefix-conditioned questions. Model outputs play no role in selection or source derangement, and the public manifests contain no answer-bearing field.

PID2Graph/OPEN100 was independently screened for the same tag-retrieval task. The official 9,303,633,645-byte ZIP, published with MD5 `90f782220de97e7e249d2595c49ddc1c`, was accessed through a sparse byte-range reader. Its central directory contains 74,655 members. All 24 image/GraphML members for 12 complete OPEN100 plans were materialized and hashed. Across these files, the GraphML provides 4,629 node class labels, 4,567 edge class labels, and node bounding boxes, but no explicitly named text, tag, or identifier data field. Internal graph node IDs are implementation identifiers and are never treated as visible tag references. The automated task-fit decision is therefore to retain PID2Graph as a structural resource without reporting a tag-retrieval score. This is a reference-availability boundary, not an acquisition failure.

S13. Qualified-budget robustness, cross-model recovery, and DEXPI external results

The Qwen robustness extension contains 24 complete cells and 7,360 immutable prediction rows: three mutually source-disjoint subsets, four image-quality conditions, and correct/source-shuffled pairing. Every row has status `ok`, `test_answer_used=false`, and the frozen-plan SHA-256 recorded in the machine-readable score report, a 3072-pixel maximum side, a 512-token cap, and 35,979,264 recorded processor tensor elements. No degraded condition changes source membership, question, prompt, answer-hidden contract, or shuffled assignment.

Pooled over 230 sources, clean/JPEG-Q70/blur-radius-1/0.75×-restore correct-image F1 values are 0.5693/0.5836/0.5879/0.5605. Corresponding source-shuffled values are 0.0114/0.0114/0.0125/0.0123. The four paired correct-minus-shuffled effects are +0.5578 ([0.4917, 0.6227]), +0.5722 ([0.5060, 0.6344]), +0.5754 ([0.5101, 0.6356]), and +0.5482 ([0.4763, 0.6134]). Degraded-minus-clean changes are +0.0143 ([-0.0034, 0.0345]), +0.0176 ([-0.0155, 0.0517]), and -0.0096 ([-0.0655, 0.0520]). These are paired source-bootstrap intervals with 10,000 replicates.

Table 18: Correct-minus-source-shuffled strict tag F1 by source-disjoint subset and quality condition. Brackets are 95% paired source-bootstrap intervals.

Subset	Clean	JPEG Q70	Blur $r = 1$	0.75× restore
Set B (100)	0.5319 [0.4252, 0.6374]	0.5357 [0.4309, 0.6397]	0.5656 [0.4543, 0.6707]	0.5175 [0.3991, 0.6281]
Strict seed 29 (65)	0.5862 [0.4949, 0.6763]	0.5976 [0.5042, 0.6900]	0.5713 [0.4891, 0.6520]	0.5915 [0.5119, 0.6694]
Strict seed 31 (65)	0.5721 [0.4650, 0.6767]	0.6097 [0.5068, 0.7028]	0.5997 [0.4977, 0.6909]	0.5563 [0.4516, 0.6643]

The closest-safe InternVL extension contains nine complete cells and 2,760 immutable prediction rows: the same three mutually source-disjoint subsets crossed with correct, source-shuffled, and no-image conditions. Every row has status `ok`, `test_answer_used=false`, and the frozen-plan SHA-256 recorded in the machine-readable score report. Image rows record 54 non-overlapping 448-pixel tiles, no thumbnail, a 512-token cap, and 32,514,048 input tensor elements; no-image rows contain no pixel tensor.

Pooled correct/source-shuffled/no-image F1 values are 0.4846/0.0131/0. The correct-minus-shuffled effect is +0.4715 [0.4064, 0.5332], and the correct-minus-no-image effect is +0.4846 [0.4198, 0.5463]. The paired 54-tile minus native-12 correct-image gain is +0.4681 [0.4043, 0.5290]. All three subset-specific correct-minus-control and 54-minus-native intervals have positive lower bounds.

Table 19: Closest-safe InternVL3.5-8B visual-budget comparison. The 54-tile input has 32.51 million tensor elements versus 35.98 million for the qualified Qwen processor; this does not imply encoder equivalence. Intervals use 10,000 paired source-bootstrap replicates.

Subset	Correct F1	Shuffled F1	No-image F1	Correct−shuffled [95% interval]	54-tile−native-12 [95% interval]
Set B (100)	0.4584	0.0033	0.0000	+0.4551 [+0.3520, +0.5581]	+0.4458 [+0.3440, +0.5500]
Seed 29 disjoint (65)	0.5087	0.0155	0.0000	+0.4932 [+0.4062, +0.5759]	+0.4964 [+0.4188, +0.5756]
Seed 31 disjoint (65)	0.5110	0.0258	0.0000	+0.4851 [+0.3943, +0.5768]	+0.4853 [+0.3924, +0.5773]
Pooled (230)	0.4846	0.0131	0.0000	+0.4715 [+0.4064, +0.5332]	+0.4681 [+0.4043, +0.5290]

The DEXPI Qwen matrix contains 195 immutable rows (65 per correct, cross-case shuffled, and no-image condition); the frozen OCR file adds 65 rows. All rows pass status, plan-hash, membership, and answer-use checks. Correct-image Qwen has TP/FP/FN 72/6/9, precision 0.9231, recall 0.8889, F1 0.9057, exact-set accuracy 0.8615, and logical-case-macro exact accuracy 0.9006. Shuffled Qwen has TP/FP/FN 0/144/81 and F1 0; no-image Qwen has 0/0/81 and F1 0. PaddleOCR has 49/2/32, precision 0.9608, recall 0.6049, and F1 0.7424. Ten-thousand- replicate logical-case-bootstrap differences are +0.9057 [0.8409, 0.9841] for correct minus shuffled, +0.9057 [0.8431, 0.9844] for correct minus no image, +0.1632 [0.0317, 0.3882] for correct minus OCR, and +0.7424 [0.5435, 0.8504] for OCR minus no image.